

Inventory Systems for Dependent Demand

When we are dealing with end products, such as automobiles, refrigerators, and personal computers, the inventory models presented in the previous section are often applicable. This is especially true when the demand for these types of products is considered to be independent and relatively constant throughout the year. However, there is another category of products: the subassemblies and components that go into these end products. For this group of products, the demand is not independent, but rather, it is dependent on the demand for the end products in which they are used. For example, the demand for automobile tires is dependent on the demand for cars. Similarly, the demand for keyboards is dependent on the demand for personal computers. In both cases, once the demand for the end product has been established, the demand for the components used in these products can be determined easily. To address the inventory issues that are associated with these types of products, a concept known as **Materials Requirements Planning (MRP)** is used.

Before directly going to MRP, let's first see an overview of two plans that should be developed prior to MRP: **Aggregate Production Plan (or Aggregate Plan)** and **Master Production Schedule (or Master Schedule)**. The development of viable aggregate plan and a coordinated master production schedule is the starting point for the use of a detailed computer-based planning system.

Aggregate planning is the process of planning the quantity and timing of output over the intermediate time horizon (often 3 months to 1 year). It focuses on a group of similar products/services or product line. The main purpose of the aggregate plan is to specify that combination of production rate, workforce level, and the resulting inventory on hand or backlog that both minimizes costs (efficiency) and satisfies the forecasted demand (effectiveness). Production rate refers to the quantity of product completed per unit of time (such as automobiles per day). Workforce level is the number of workers needed for production. When the number of units produced in any given period exceeds demand, the result is an inventory on hand of the product. When demand exceeds production, the result is a backlog (or stock out), which represents the shortfall. Both inventories and backlogs are carried forward to the next time period. However, there can be situations when stock outs are not carried forward because the customer decided to purchase the product elsewhere rather than wait.

The master production schedule (MPS) is developed directly from the aggregate plan-and is the instrument that drives the firm's entire production system. The aggregate plan establishes an overall level of operations that balances the plant's capability with external sales demand. The master schedule translates the aggregate plan into specific numbers of specific products to be produced in identified time periods.

The relationship between the master schedule and the aggregate plan is shown in Figure 4.3. In this hypothetical illustration, an appliance manufacturer's aggregate plan for refrigerator production is shown at the top of the figure. Note that the master production schedule breaks down the aggregate production of refrigerators into the production of seven specific models, time-phased by quantity per month for the planning period of a year. For example, the 500 units planned in January consist of 200 Model A standards, 50 Model A Deluxe units, 200 Model B Standards, and 50 Model B Deluxe units. The master schedule for the rest of the year was constructed in the same manner.

Aggregate plan for Refrigerators

Month	Jan.	Feb.	Mar.	Apr	May	June	July	Aug.	Sept.	Oct	Nov.	Dec.
Number of refrigerators	500	500	600	700	700	800	800	700	600	500	400	400

Master Production Schedule for Refrigerators

Month	Jan.	Feb.	Mar	Apr.	May	June	July	Aug	Sept.	Oct	Nov.	Dec.
Business Model												
Standard	-	100	-	200	-	200	-	150	-	100	-	100
Heavy duty	-	100	-	100	-	100	-	100	-	50	-	50
Model A												
Standard	200	100	300	150	300	200	350	200	250	150	150	100
Deluxe	50	-	50	-	100	-	100	-	50	-	50	-
Executive	-	100	-	150	-	200	-	150	-	100	-	100
Model B												
Standard	200	100	200	100	200	100	250	100	250	100	150	50
Deluxe	50	-	50	-	100	-	100	-	50	-	50	-

Figure 1: Illustrative relationships of the aggregate plan and the master production schedule.

Material Requirements Planning (MRP)

MRP is a computer-based information system that translates the finished product requirements of the master schedule into time-phased requirements for subassemblies, component parts, and raw materials, working backward from the due date using lead times and other information to determine when and how much to order. Material requirements planning is as much a philosophy as it is a technique, and as much an approach to scheduling as it is to inventory control. MRP is designed to answer three questions: what is needed? how much is needed? and when it is needed?

MRP Inputs

An MRP system has **three major sources of information**: a master schedule, a bill-of-materials file, and an inventory records file. Let's consider each of these inputs.

The Master Schedule

The master schedule, also referred to as the master production schedule, states which end items are to be produced, when they are needed, and in what quantities. The quantities in a master schedule come from a number of different sources, including customer orders, forecasts, and orders from warehouses to build up seasonal inventories.

The Bill of Materials

A bill of materials (BOM) contains a listing of all of the assemblies, subassemblies, parts, and raw materials that are needed to produce one unit of a finished product. Thus, each finished product has its own bill of materials.

The listing in the bill of materials is hierarchical; it shows the quantity of each item needed to complete one unit of the following level of assembly. The nature of this aspect of a bill of materials is clear when you consider a product structure tree, which provides a visual depiction of the subassemblies and components needed to assemble a product.

A product structure tree is useful in illustrating how the bill of materials is used to determine the quantities of each of the ingredients (requirements) needed to obtain a desired number of end items.

Level

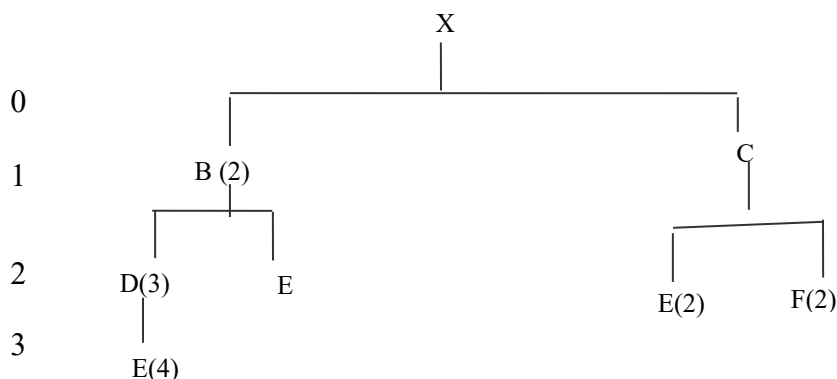


Figure 2: A Product Structure Tree for end item X.

Let's consider the product structure tree shown in Figure 2. End item X is composed of two Bs and one C. Moreover, each B requires three Ds and one E, and each D requires four Es. Similarly, each C is made up of two Es and two Fs. These requirements are listed by level, beginning with 0 for the end item, then 1 for the next level, and so on. The items at each level are components of the next level up and, as in a family tree, are parents of their respective components. Note that the quantities of each item in the product structure tree refer only to the amounts needed to complete the assembly at the next higher level.

Exercise: Use the information presented in figure 2 to do the following:

- Determine the quantities of B, C, D, E, and F needed to assemble one X.
- Determine the quantities of these components that will be required to assemble 10 Xs, taking into account the quantities on hand (i.e., in inventory of various components):

Component	On hand
B	4
C	10
D	8
E	60

The Inventory Records

Inventory records refer to stored information on the status of each item by time period, called time buckets. This includes gross requirements, scheduled receipts, and expected amount on hand. It also includes other details for each item, such as supplier, lead time, and lot size policy changes due to stock receipts and withdrawals, canceled orders, and similar events also are recorded in this file.

Like the bill of materials, inventory records must be accurate. Erroneous information on requirements or lead times can have a detrimental impact on MRP and create turmoil when incorrect quantities are on hand or expected delivery times are not met.

MRP Processing

The essence of MRP is determining the quantity and timing necessary for each component in order to achieve the quantity and timing of end items in the master schedule. The process then begins with the master schedule. Then, each end item is “exploded” in to its component parts using the bill of materials file, indicating the assemblies, sub-assemblies, parts and raw materials that will be needed. Conceptually, this is the same as identifying an end item's product tree. Next, the actual amounts (net requirements) of each component are determined, level-by-level, beginning at the top of the tree and working down. This is the heart of MRP; it involves determining the quantity of each item called for (gross requirements) and, then, subtracting out inventory available, which consists of any inventory on hand from previous period plus orders scheduled to be available (schedule receipts).

Net requirements are deemed planned order receipts until the order are actually executed, at which time they become schedule receipts. In order to meet the quantity and timing of net requirement, orders are planned for execution (planned order release) prior to planned order receipt according to the lead-time needed. The computation of net requirements of each component is based on this formula:

$$\begin{array}{ccccccc} \text{Net} & & \text{Gross} & & \text{Projected} & & \\ \text{requirements} & = & \text{requirements} & - & \text{inventory} & + & \text{Safety stock} \\ \text{in period } t & & \text{in period } t & & \text{in period } t & & \text{(if any)} \end{array}$$

Definition of Terms

Gross requirement: The total expected demand for an item or raw materials during each time period without regard to the amount on hand. For end items, these quantities are shown in the master schedule; for components, these quantities are derived from the planned-order releases of their immediate “parents”.

Scheduled receipts: Open orders (orders that have been placed and are) scheduled to arrive from vendors or elsewhere in the pipeline by the beginning of a period.

Projected on hand: The expected amount of inventory that will be on hand at the beginning of each time period: scheduled receipts plus available inventory from last period.

Net requirements: The actual amount needed in each time period.

Planned-order receipts: The quantity expected to be received by the beginning of the period in which it is shown. Under lot-for-lot ordering, this quantity will equal net requirements. Under lot-size ordering, this quantity may exceed net requirements. Any excess is added to available inventory in the next time period for simplicity, although in reality, it would be available in that period.

Planned-order releases: Indicates a planned amount to order in each time period; equals planned-order receipts offset by lead time. This amount generates gross requirements at the next level in the assembly or production chain. When an order is executed, it is removed from “planned-order releases” and entered under “scheduled receipts.”

These quantities are used in a time phased plan in the following format. The column for week 0 is used to show beginning inventory.

Week Number	0	1	2	3	4	5	6	7	8
-------------	---	---	---	---	---	---	---	---	---

Item:									
Gross requirements									
Scheduled receipts									
Projected on hand									
Net requirements									
Planned-order receipts									
Planned-order releases									

Lot Sizing: Determining a lot size to order or to produce is an important issue in inventory management for both independent-and dependent demand items. This is called lot sizing. For independent demand items, managers often use economic order sizes and economic production quantities. For dependent-demand systems, however, a much wider variety of plans is used to determine lot sizes, mainly because no single plan has a clear advantage over the others. Not surprisingly, the methods used to handle lot sizing range from the complex, which attempt to include all relevant costs, to the very simple, which are easy to use and understand. In certain cases, the simple models seem to approach cost minimization although generalizations are difficult. Let's consider some of these models.

Lot-for-Lot Ordering: Perhaps the simplest of all the methods is lot-for-lot ordering. The order or run size for each period is set equal to demand for that period. Not only is the order size obvious, it also virtually eliminates holding costs for parts carried over to other periods. Hence, lot-for-lot ordering minimizes investment in inventory. Its two drawbacks are that it usually involves many different order sizes and thus cannot take advantage of the economics of fixed order size, and it requires a new setup for each production run. If setup costs can be significantly reduced this method may approximate a minimum-cost lot size.

Economic Order Quantity Model: Sometimes economic order quantity (EOQ) models are used. They can lead to minimum costs if usage is fairly uniform. This is sometimes the case for lower-level items that are common to different parents and for raw materials. However, the more lumpy demand is the less appropriate such an approach is. Since demand tends to be most lumpy at the end item level, EOQ models tend to be less useful for end items than for items and materials at the lowest levels.

Fixed-period Ordering: This type of ordering provides coverage for some predetermined number of periods (e.g., two or three). In some instances, the span is simply arbitrary, in other cases, a review of historical demand patterns may lead to a more rational designation of a fixed period length. A simple rule is order to cover a two-period interval. The rule can be modified when common sense suggests a better way.

JIT and Lean Operations (independent Review)

The term just-in-time (JIT) is used to refer to an operations system in which materials are moved through the system, and services are delivered with precise timing so that they are delivered at each step of the process just as they are needed-hence the name just-in-time. Initially, the term JIT referred to the movement of materials, parts, and semi-finished goods within a production system. Over time, the scope of JIT broadened and the term became associated with lean production. Now the two terms are often used interchangeably to refer to a highly coordinated, repetitive manufacturing or service system designed to produce a high volume of output with fewer resources than more traditional repetitive systems, but with ability to accommodate more variety than traditional systems.

One aspect of JIT involves production on planning and control, which makes JIT one of the two basic approaches to manufacturing planning and control; the other is material requirements planning. JIT is sometimes viewed as a system for repetitive production operations, while MRP is considered a system for batch production. However, they are sometimes applied to similar situations, but the two systems function somewhat differently. MRP systems are fairly complex, requiring extensive and detailed shop floor controls. JIT systems are much simpler, involving only minimal shop floor controls. Moreover, MRP relies on a computer-based component scheduling system to trigger production and deliveries, whereas JIT relies on visual or audible signals to trigger production and deliveries. Also, JIT operates on a "pull" or demand bases where by work is pulled from each step in the process to the next step when the next step has demand for it, rather than pushed on to the next step when work is completed at the current step.

The JIT approach was developed at the Toyota Motor Company of Japan by Taiichi Ohno (who eventually became vice president of manufacturing) and several of his colleagues. The development of JIT in Japan was probably influenced by Japan being a crowded country with few natural resources. Not surprisingly, the Japanese are very sensitive to waste and inefficiency. They regard scrap and rework as waste and excess inventory as an evil because it takes up space and ties up resources.

In some respects, the just-in-time concept was operational over sixty years ago at Henry Ford's great industrial complex in River Rouge, Michigan. Toyota learned a great deal from studying Ford's operations and based its JIT approach on what it saw. However, Toyota was able to accomplish something that Ford couldn't: a system that could handle variety.

A widely held view of JIT is that it is simply a system for scheduling production that results in low levels of work-in-process and inventory. But in its truest sense, JIT represents a philosophy that encompasses every aspect of the process, from design to after the sale of a product. The philosophy is to pursue a system that functions well with minimal levels of inventories, minimal waste, minimal space, and minimal transactions. truly, a lean system. As such, it must be a system that is not prone to disruptions and is flexible in terms of the product variety and range of volume that it can handle.

In lean systems, quality is ingrained in both the product and the process. Companies that use lean operations have achieved a level of quality that enables them to function with small batch sizes and tight schedules. Lean systems have high reliability; major sources of inefficiency and disruptions have been eliminated, and workers have been trained not only to function in the system but also to continuously improve it.

The ultimate goal of JIT is a balanced system, that is, one that achieves a smooth, rapid flow of materials and/or work through the system. The idea is to make the process time as short as possible by using resources in the best possible way. The degree to which the overall goal is achieved depends on how well certain supporting goals are achieved. Those goals are:

- Eliminate disruptions.
- Make the system flexible.
- Eliminate waste, especially excess inventory.

Disruptions have a negative influence on the system by upsetting the smooth flow of products through the system, and they should be eliminated. Disruptions are caused by a variety of factors, such as poor quality, equipment breakdowns, changes to the schedule, and late deliveries. All these should be eliminated as much as possible. This will reduce the uncertainty that the system must deal with.

A flexible system is one that is robust enough to handle a mix of products, often on a daily basis, and to handle changes in the level of output while still maintaining balance through-put speed. This enables the system to deal with some uncertainty. Long setup times and long lead times negatively impact the flexibility of the system. Hence, reduction of setup and lead times is very important in a JIT system.

Waste represents unproductive resources; eliminating wastes can free up resources and enhance production. Inventory is an idle resource, taking up space and adding cost to the system. It should be minimized as much as possible. In the JIT philosophy, other wastes include.

Over production – involves excessive use of manufacturing resources.

Waiting time – requires space, adds no value.

Unnecessary transporting – increases handling, increase work-in-process inventory.

Processing waste- makes unnecessary production steps, scrap.

Inefficient work methods – reduce productivity, increase scrap, increase work-in-processes inventory.

Product defects – require rework costs and possible lost sales due to consumer dissatisfaction.

The existence of these wastes is an indication that improvement is possible. The list of wastes also can identify potential targets for continuous improvement efforts.

The design and operation of a JIT system provide the foundation for accomplishing the aforementioned goals. The building blocks are:

Product design four elements of product design are important for JIT: standard parts, modular design, highly capable production systems with quality built in and concurrent engineering.

Process design seven aspects of process design are particularly important for JIT systems: small lot sizes, setup time reduction, manufacturing cells, limited work in process, quality improvement, production flexibility and little inventory storage.

Personnel/Organizational elements: There are five elements of personnel and organization that are particularly important for JIT systems: workers as assets, cross-trained workers, continuous improvement, cost accounting and leadership/project management.

Manufacturing Planning and Control Six elements of manufacturing planning and control are particularly important for JIT systems: level loading, pull systems, visual systems close vendor relationships, reduced transaction processing, and preventive maintenance and housekeeping. Speed and simplicity are two common threads that run through these building blocks.

In summary, the following elements tend to characterize most successful JIT operations.

The JIT concept is most applicable to manufacturing operations that produce a relatively small number of different products in at least a quasi-continuous environment.

Product demand must be reasonably predictable, and requirements must be generated accurately. A closed loop MRP system can be used to do this, but typically the master production schedule must be smoothed on a daily basis.

Statistical process control typically is used in both the buyer's and suppliers' organizations to ensure tight control of material and production quality. This is vital to the functioning of the low-float, small-volume, relatively smooth-flowing operation.

Production operation setup requirements must be able to be reduced to relatively short times. Most firms target for tool changes and equipment setups of less than ten minutes. Without this capability, small-batch and smooth-flow production of different model of different products cannot be accomplished efficiently.

Purchasing must be able to reduce materials replenishment lead times. This usually is accomplished by reducing the four major elements of lead time-internal paperwork and ordering time, supplier queue and manufacturing time, transportation time requirements, and incoming receiving and inspection requirements.

Successful JIT operation suppliers must be able to be flexible to meet the buying firm's stringent, short-fused material requirements and they must be reliable.

Inventory Classification Systems

Any organization has to stock and keep track of large number of items of different kinds. These may include raw material supplies for production process, spare parts and maintenance items for various other purposes. As soon as an inventory is identified and described, the manager must determine the importance and the dollar value of each individual inventory item. Importance of an item largely arises due to the two factors:-

- Consumption value and criticality in use: The production or operation manager should recognize the unequal contribution of different items in the inventory. Equal effort may not be spent in managing all inventories.
- For effective inventory control: decision has to be made that which items are little things and which need more careful control.

Inventories can be classified into various groups on the basis of the selective inventory management approach as follows.

1. ABC Inventory Analysis

ABC analysis is a basic analytical management tool which enables top management to place the effort where the results will be greatest. This technique, popularly known as Always Better Control or the alphabetical approach, has universal applications in many areas of human endeavor. The technique tries to analyze the distribution of any characteristic **by money value** of importance in order to determine its priority. In materials management, this technique has been applied in areas needing selective control, such as inventory, criticality of items, obsolete stocks, purchasing orders, receipt of materials, inspection, store keeping and verification of bills.

Even though there is lack of clear-cut principle to classify items in to A, B and C for all organizations, the normal items in most organizations show the following pattern:

5 % to 10 % of the top number of items account for about 70 % to 80 % of the total consumption value. These items are called 'A' items.

10 % to 20 % of the number of items account for 10 % to 15 % of the total consumption value. These items are called 'B' items.

The remaining number of items accounts for the balance 65 % to 80 % of the number of items and about 5 % to 10 % of the total consumption value. These items are called 'C' items.

ABC procedure

The mechanics of classifying the items into 'A', 'B' and 'C' categories is described as follows

- Calculate annual usage in units for each item for the past one year.
- Calculate the annual usage in birr for each item by multiplying the annual usage quantity with the average unit price of each item.
- Rank the items from highest birr usage annually to the lowest annual usage in birr.
- Arrange the items in the inventory by cumulative annual usage (birr) and by cumulative percentage.
- Categorize the items in A, B, and C categories

Control through ABC Analysis

- Top level of management must devote a major part of its time in procurement of 'A' category items, both from the point of view of cost control as well as controlling the quantity to be procured.
- 'B' category items also deserve attention from top level of management, but not to the same extent as that of an 'A' category item. The middle level of management may be given the responsibility to monitor the procurement of this category of items. The top level of management should just oversee that actions for procurement of the 'B' category are taken as per directions and guidelines laid down by it.
- The lower level of management should manage the 'C' category items.

Example1:

XYZ factory adopts the ABC method of classifying inventories. Currently, the factory has 10 items. The following is the data related to the items.

Item No	Annual usage, Q	Unit cost (birr)
22	1100	2
68	600	40
27	100	4
03	1300	1
82	100	60
54	10	25
36	100	2
19	1500	2
23	200	2
41	500	2

Classify the items into ABC with A items taking 80%, B items taking about 15% and C taking 5% of the total birr value.

Example 2:

Classify the following items into A, B, and C

Item	Annual consumption	Unit Price
K	7,000	0.10
L	2,000	0.70
M	10,000	0.02
N	4,000	0.05
O	1,000	0.30
P	10,000	0.60
Q	7,000	0.02
R	2,000	0.03

3. VED (Vital, Essential, Desirable) Analysis

This analysis is based on the criticality of inventory.

V-item – are items when go out of stock or when not readily available, completely bring the production to a halt. So, they should be stored adequately to insure continuous production.

E-item – are items without which temporary losses of production or dislocation of production work occurs.

D-item – are all other items which are necessary but do not cause any immediate effect on production.

4. SDE (Scarce, Difficult, Easily) Analysis

This analysis is based on availability of items (raw materials).

S-item – are items which are in small supply and are usually imported items.

D-item – are items which are available in the market but cannot be procured easily. For example, items which have to come from far off cities.

E-item – are easily available items; mostly local items

5. HML (High, Medium, Low) Analysis

The cost per item is considered for this analysis. High cost item (H), Medium cost items (M) and Low cost items (L) help in bringing control over consumption at the department level.

6. FSN (Fast, Slow, Normal) Analysis

Here the quantity and rate of consumption is analyzed to be classified as fast moving, normal, and slow. This classification helps in arranging stocks in the stores according to the frequency that the items are used or consumed.

7. XYZ Analysis: The analysis is based on the value of closing inventory.

X-items – Items with high closing inventory.

Y-items – Items with moderate closing inventory.

Z-items – Items with low closing inventory.